Applied Social Data Science

Coding Camp

Setup · Orientation

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Introductions

About me

- Hey, I am Elena!
- PhD student in Political Science
- Research on blame attribution after natural disasters and implications on voting and policy change
- Main methods I use in my dissertation: computational methods (LLMs), quantitative methods, quasi-experiments
- Coding: Python, R, LaTeX

About me

Blame and Gain: Does Blame Attribution after a Natural Disaster Increase Support for the Incumbent? A German case study

Elena Karagianni

Trinity College Dublin

Abstract

Do the words politicians choose after a natural disaster move voters, independently of how badly the disaster hit or how well the government responded? I study this question in the German federal setting, where the same hazard routinely crosses state borders but crisis management and official communication remain the responsibility of each *Land*. This institutional feature generates near-identical shocks met by governments that talk about them differently. I make two contributions. First, I use a three-pass large language model (LLM) pipeline to identify disaster-related parliamentary speeches and classify each incumbent party's blame-attribution strategy — internalizing blame, avoiding (externalizing) blame, credit-claiming, proposing policy, and neutral — across a corpus of 36,599 state-parliament speeches (2002–2022). Second, I estimate the electoral effect of these strategies with a stacked, cross-state difference-in-differences design that conditions on objective government response, and that reads a causal signal off an *ambiguity gradient*. Across the nine identifying disasters, governments that meet a disaster with blame avoidance lose poll support, an effect concentrated among *junior* coalition partners.

Local Moves, Big Gains? Examining the Party Activity Strategies of the Far-Right

Constantine Boussalis[1], Elena Karagianni[1], and Lamprini Rori[2]

[1]Trinity College Dublin
[2]National and Kapodistrian University of Athens

Abstract

How do far-right parties convert local activism into electoral support? We address this question with an original dataset of local outdoor activities organised by Golden Dawn in Greece between 2012 and 2015. Relying on computational methods, we extract, geolocate, and classify every event reported on the party's website, then merge these data with municipal-level election returns and census indicators. Two-way fixed-effects models show that not all activism pays off. Municipalities that hosted more community engagement events (e.g., visits to local stakeholders) or ceremonial and cultural gatherings (e.g., nationalist commemorations) subsequently gave Golden Dawn higher vote shares. By contrast, the party's highly publicised "Greeks-only" welfare distributions and confrontational street mobilisations provided no electoral benefit; and in the former case, may have even reduced support. Our findings demonstrate that the type of far-right activism matters and highlight the normalising power of community outreach activities and identity affirming rituals.

Introductions

https://www.menti.com/alxfwcooky1x

Schedule

10am - 1pm

- **Monday: Introduction + R Basics I**
- Tuesday: R Basics II + Good practices

10am - 12pm

- Python basics
- Friday: How to write up and report on LaTeX

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Used for:

- Better decisions
- Predictive analysis
- Pattern discovery

A brief history

John Tukey, 1915–2000

'All in all, I have come to feel that my central interest is in **data analysis**, *which I take to include, among other things: procedures for analyzing data, techniques for interpreting the results of such procedures, ways of planning the gathering of data to make its analysis easier, more precise or more accurate, and all the machinery and results of (mathematical) statistics which apply to analyzing data.'*

The Future of Data Analysis, 1962.

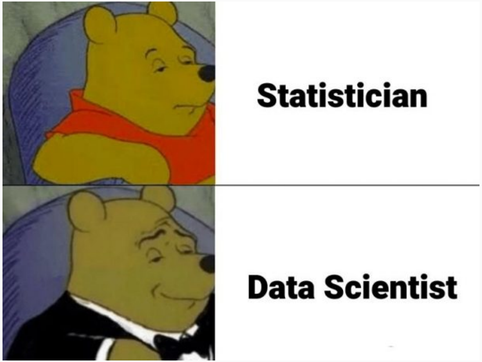
Things to keep in mind

Key distinction

Statistics = mathematics of inference

Data science = practice of working with data

- Not everyone agrees with this distinction
- In practice, the terms are often used differently



Things to keep in mind

- There is a difference between **scientific** and **engineering** mindsets
  - **Scientific mindset** $\rightarrow$ seeks to understand the underlying process (generative modeling)
  - **Engineering mindset** $\rightarrow$ looks to find the best prediction (predictive modeling)
- In the social sciences, we often want to understand what's inside the 'black box', but not all data science methods are designed for this.

Things to keep in mind

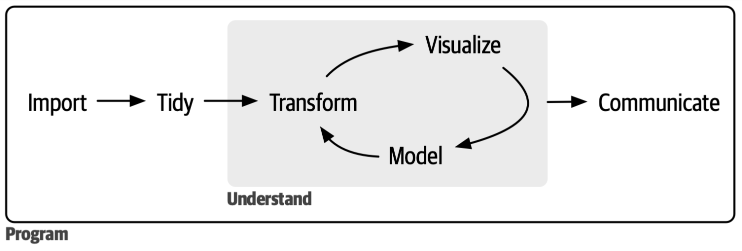
Data Science

- Broad field: principles & methods to identify and understand phenomena
- Uses programming, statistics, **machine learning**, and algorithms
- Works with large datasets to discover patterns and predict outcomes

Data Analytics

- Focused on insights for decision-making
- Analyzes past data to answer specific questions
- Supports present decisions with evidence

Data Science is a Process



Introduction, R for Data Science, c/o Hadley Wickham

80 % of the data analysis process is spent on the process of cleaning and preparing the data!

R and Python


YES, I WOULD LOVE TO SEE ANOTHER
R VS PYTHON FOR DATA SCIENCE COMPARISON
makeameme.org

Installing R

R is a programming language used for statistics. It is completely free and can be downloaded from **CRAN**, the comprehensive R archive network.

1. Go to `https://cran.r-project.org/`
2. In the box headed “Download and Install R”, click the link corresponding to your operating system.
3. Follow the instructions for your system.

Installing **R Studio** $\rightarrow$ integrated development environment

1. Go to `https://docs.posit.co/ide/user/#rstudio-ide-oss-downloads`
2. Install the version corresponding to your operating system.
3. Follow the instructions for your system.

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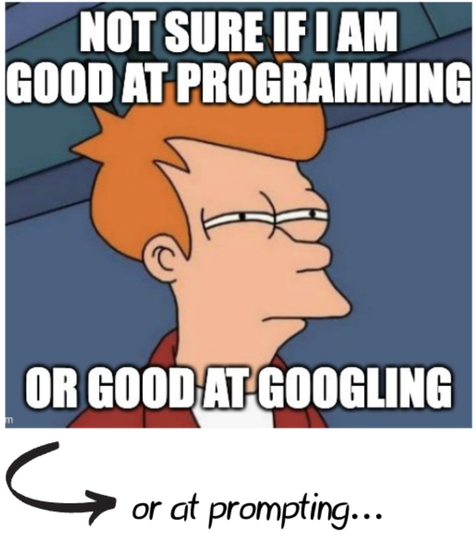
- If you are already familiar with all this, try helping the person next to you!



Useful Resources

- R for Data Science (2e): `https://r4ds.hadley.nz/`
- Python for Data Analysis (3e): `https://wesmckinney.com/book/`
- GitHub: `https://docs.github.com/en/get-started/quickstart/hello-world`
- Posit Recipes: `https://posit.cloud/learn/primers`

ASDS GitHub

`https://github.com/ASDS-TCD/Code_Camp_2026`